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SIREX WOODWASP



Amylostereum complex – Sirex noctilio F.

Last updated by:
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Updated 2022 Overview and Introduction to North America

The woodwasp, *Sirex noctilio*, is native to Eurasia and North Africa (Hajek, Haavik, and Stephen, 2021*). Its associated fungus, *Amylostereum areolatum*, has numerous strains native to much of the Northern hemisphere including North America, Eurasia, and North Africa.

In North America, *S. noctilio* had been repeatedly intercepted in wood packaging before implementation of the international standard (ISPM#15). Over the period 1993 – 2001, USDA inspectors found 103 shipments containing wood packaging contaminated with *S.*

noctilio. The site where the woodwasp was initially detected is along the St. Lawrence Seaway, a major route for imports. Analysis of the woodwasp's and the fungus' genetic patterns indicates that *S. noctilio* was introduced at least twice to North America – once from an unknown location in Europe and once from South America. The woodwasp was detected in a trap (part of the USDA/CAPS woodborer trapping program) in Oswego, New York near Lake Ontario in 2004. Subsequent delimitation surveys revealed that the woodwasp was present and had persisted for many years; eradication was thus considered infeasible. The woodwasp and multiple strains of the fungus were also introduced to New Zealand, Australia, South America, and South Africa during the 20th Century.

By 2020, the woodwasp and associated fungi were found in nine Northeastern states (New York, Pennsylvania, Vermont, Connecticut, Ohio, Michigan, New Jersey, Massachusetts and New Hampshire) and two Canadian provinces (Ontario and Quebec). While a coordinated survey effort is lacking in both the United States and Canada, these pests appear to be spreading.

Following the initial detection, the US Forest Service funded work on the fundamental biology and ecology of native North American Siricidae in order to form a baseline knowledge set before *S. noctilio* could significantly impact the environment. This included an economic risk assessment (USDA Forest Service, 2006), whereby researchers projected that the woodwasp could spread across the entire southern pine region in 55 years or less, with damage ranging from \$2 billion to \$11 billion.

Impacts in Northeastern North America

There are no long-term data on *Sirex noctilio*'s impact on pine forests across the northeastern United States (or, apparently, eastern Canada). Attempting to measure such impacts proves difficult because 1) determining whether a tree's death was caused by the woodwasp must occur within a short period of time, 2) finding infestations is impeded by the lack of a good lure, and 3) highly fragmented pine stands limit comparative studies. What is clear, however, is that in northeastern North America, the wasp feeds primarily on the native red (*Pinus resinosa*) and introduced Scots pines (*P. sylvestris*). In contrast, the native jack pine (*P. banksiana*) is attacked with low frequency, and the eastern white pine (*P. strobus*) is rarely attacked. These pines grow primarily in widely scattered stands, most of which were planted 70+ years in the past with limited management today. In all studies, pine mortality has often been restricted to suppressed and intermediate crown classes and damaged or dying trees; however, the small sizes of the studies and difficulty in comparing study results from stands with varied conditions obscures efforts to determine whether impacts are driven primarily by a host species' vulnerability or by stand conditions.

Pine mortality is greatest early in the invasion cycle, then declines to a low, background level. This reduction in pine mortality rate likely follows exhaustion of susceptible host material; it might also reflect responses by local parasitoids after a time lag.

In the northeast region of North America, the woodwasp appears to be behaving largely as it does in Eurasia, feeding on weakened or dying pines growing at high densities. This contrasts with its behavior in the introduced range in the Southern Hemisphere where it attacks plantations of non-native pines planted at high densities. Under the latter condition, *S. noctilio* can reach epidemic densities and cause considerable tree mortality. Low host availability in North America (NA) might explain the inability of the woodwasp to amass damaging population levels.

Biology/life cycle

Like other xylem-chewing insects (e.g., ambrosia beetles), siricids have co-evolved with mutualistic fungi that break down this material into compounds that can be readily assimilated. See Hajek, Haavik and Stephen (2021) for a full description of the complicated

Sirex/Amelosterium pest complex life cycle.

Interactions with other forest insects

A native congener, *Sirex nigricornis*, is found across eastern North America and as far west as Alberta. It is the only native woodwasp in eastern North America that uses pine as its exclusive host. It colonizes only severely weakened trees. While its range overlaps the invaded range of *S. noctilio* (they can be found using the same host tree) in almost all instances of co-colonization, the non-native woodwasp greatly outnumbers *S. nigricornis*. Future interactions between the two woodwasps are difficult to predict. *S. noctilio* is now the most abundant siricid colonizing pines in the northeast region of NA and has almost certainly impacted native insect communities. In the absence of data, however, it is unknown whether *S. noctilio* has negatively affected native siricids with a competing population density, or positively affected them by created an abundance of weakened host trees that native siricids can exploit.

Within the invaded North American range, several native hymenopteran parasitoids parasitize up to 25% of *S. noctilio* larvae (see Hajek, Haavik and Stephen, 2021 Chapters 6, 7, and 8). All are solitary parasitoids, meaning that one *S. noctilio* larva yields one parasitoid. These parasitoids do not appear to be a major regulator of the woodwasp's population dynamics.

Limited research exists on the impact of other boring insects on *S. noctilio* in North America. However, native and introduced bark beetles that attack weakened pines carry their own fungal symbionts (e.g., the bluestain fungus *Ophiostoma* spp.). These fungi often outcompete and suppress the siricid fungus *Amylostereum areolatum* on which *S. noctilio* larvae depend for feeding. Researching this issue presents a challenge since the same tree can harbor multiple cohorts of different siricids and different parasitoids, all of them impossible to identify as larvae. It is clear, however, that North American native hymenopteran parasitoids have readily adopted *S. noctilio*. As noted above, levels of parasitism reach 20%.

Management

Foresters in the Southern Hemisphere have learned to reduce damage caused by the woodwasp/Amylostereum complex by thinning trees to reduce density and introducing a parasitic nematode *Deladenus* (formerly *Beddingia*) *siricidicola*. One strain of this nematode, the Kamona strain, sterilizes the female woodwasps, thus reducing woodwasp populations and impacts. Some countries also use classical biocontrol relying on parasitic wasps. Success of these management strategies varies among the invaded countries (see Hajek, Haavik and Stephen, 2021 for details).

Unfortunately, the strain of *Deladenus siricidicola* nematode introduced inadvertently to North America along with the woodwasp does not sterilize the introduced woodwasp. This "introduced to North America strain", or INA strain, does parasitize a native North American beetle, *Serropalpus substriatus*. These factors have impeded efforts to manage *S. noctilio* in North America. There are, however, five species of *Deladenus* associated with native siricids and their fungal symbionts. Hajek and Morris (Chapter 7) suggest that North American foresters should explore use of a native nematode, congeneric *Deladenus proximus*, as a biocontrol. According to Hartshorn (Chapter 9), *D. proximus* has been inconsistent in regulating *S. noctilio*.

While the INA strain of the *Deladenus siricidicola* nematode does not sterilize *S. noctilio* females, it does have a negative impact on the woodwasp because parasitized adults are somewhat smaller (likely flying shorter distances) and carry fewer eggs.

Unclear danger to other parts of North America

Upon the initial detection, APHIS sought support from the softwood lumber industry for regulations. The agency issued a pest risk analysis in July 2006 (Borchert, 2006) and sponsored two scientific workshops and briefing for stakeholders (in January 2007 and May 2011). As previously mentioned, the Forest Service issued an economic analysis (USDA Forest Service, 2006) which predicted that the woodwasp could spread across the entire southern pine region in 55 years or less, with damage ranging from \$2 billion to \$11 billion; however, no consensus was built for regulations. Some scientists questioned the risk because pines native to North America have numerous other associated insects and expected that the management tool used in Southern Hemisphere countries – the parasitic nematode – could be applied. (As noted, the parasitic nematode present in North America is yet to be shown as a successful management tool.) North Carolina established an external quarantine in 2008, but other states (e.g., Georgia) did not follow suit.

While APHIS funded survey programs for several years using a small appropriation by Congress (e.g., \$15 million in FY2009), by 2021 there was no longer a systematic survey or monitoring program. The result is considerable uncertainty in baseline knowledge such as where the complex is established.

Hajek, Haavik and Stephen (2021) call on forest managers to monitor voluntarily for *S. noctilio* presence and population growth, especially in pine stands that are near thinning harvest dates, and during and after drought.

Presently, the apparent spread of the pest complex is not as fast as initially feared and has had little documented impact. Nevertheless, under the assumption that the woodwasp continues to spread, concern remains about how it will behave when it reaches parts of North America where pines dominate both natural and planted forests (i.e., the Southeastern states and the West). These forests are composed of pine species that have been killed by the woodwasp's attack in plantations in the Southern Hemisphere. In the Southeast, these include loblolly (*P. taeda*) and slash (*P. elliotii*) pines; in the West, Monterey pine (*P. radiata*), lodgepole (*P. contorta*) ponderosa (*P. ponderosa*). Mortality of these species has varied among the invaded countries. At the southern and eastern edges of its invaded range, *S. noctilio* has probably already come into contact with Virginia pine (*P. virginiana*) and pitch pine (*P. rigida*), but no attacks on these species have been reported as of 2020. Hartshorn (Chapter 9) and Foelker and Haavik (Chapters 10 & 11) discuss possible interactions between *S. noctilio* and native insect communities of the Southeast and West but are unable to reach firm conclusions.

Climatic suitability is also open to question. A climate model applied in one study found climate suitability for *S. noctilio* in the Southeast and marginal suitability in the West. The same model classified Argentine Patagonia as similarly marginal, however, this proved wrong by the damage caused by a *S. noctilio* infestation in that area. Foelker and Haavik note that overstocking and drought stress have been closely tied to *S. noctilio* population outbreaks in the Southern Hemisphere. Both conditions are common in the southeast and western regions of North America.

Tetley and Hajek (Chapter 2) conclude that the impacts of the *Sirex*/Amelosterium complex in these regions will ultimately depend on the interplay of tree species' varying susceptibility, management practices, *S. noctilio* life history traits, natural enemy and competitor communities, and landscape-level or environmental factors that influence *S. noctilio* dispersal and the availability of suitable hosts. Foelker and Haavik (Chapter 10) say scientists cannot yet specify the quantitative contributions of the three forms of biotic resistance (tree resistance, natural enemies, and competition from other pine-colonizing insects and their associated fungi).

Hajek, Haavik and Stephen (2021) do not discuss the implications for forests intended to be unmanaged, such as in National parks and Wilderness areas.

USFS scientists and managers developed a conservation priority-setting framework for forest tree species at risk from pest & pathogens and other threats. The Project CAPTURE (Conservation Assessment and Prioritization of Forest Trees Under Risk of Extirpation) uses FIA data and expert opinion to group tree species under threat by non-native pests into vulnerability classes and specify appropriate management and conservation strategies. The scientists prioritized 419 tree species native to the North American continent. The analysis identified 15 taxonomic groups requiring the most immediate conservation intervention because of the tree species' exposure to an extrinsic threat, their sensitivity to the threat, and their ability to adapt to it. Each of these 15 most vulnerable species, and several additional species, should be the focus of both a comprehensive gene conservation program and a genetic resistance screening and development effort. The *sirex* woodwasp is not known to be a threat to any of these 15 most vulnerable species.

*except when otherwise noted, material is from Hajek, Haavik, and Stephen, 2021

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Viburnum leaf beetle

Asian longhorn beetle

